

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Cycle Test

22 April 2025

Subject	Mathematics
Grade	12
Topic	Trigonometry
Marks	50
Time	1 hour

Examiner

Dr G Nixon

Moderator

Mrs S Rowell

Mrs M Trollope

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 6 questions and 4 pages. Please ensure that your question paper is complete.
2. Answer ALL the questions.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers to TWO decimal places, unless stated otherwise.
8. Diagrams are NOT necessarily drawn to scale.
9. Use the mark allocation to guide the length of your solution.
10. Write neatly and legibly.
11. Rule off after each question.

FORMULAE

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

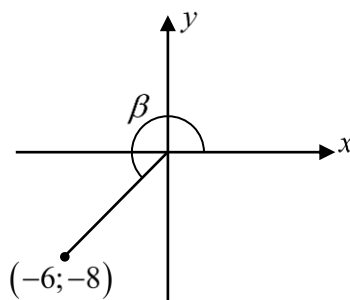
$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \\ 1 - 2 \sin^2 \alpha \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

QUESTION 1

Given that $-13\cos \alpha = 12$ and $\sin \alpha > 0$, together with the sketch below of angle β :



Determine, without the use of a calculator, the value of $\sin(\alpha - \beta)$.

[6]

QUESTION 2

Calculate, without the use of a calculator:

2.1 $\sin^2 15^\circ - \cos^2 15^\circ$ (3)

2.2 $\sin 140^\circ \cos 340^\circ - \cos 40^\circ \sin(-20^\circ)$ (5)

[8]

QUESTION 3

If $\cos 43^\circ = m$, express the following trigonometric ratios in terms of m :

3.1 $\tan 137^\circ$ (3)

3.2 $\sin 86^\circ$
(2)

3.3 $\cos 17^\circ$ (3)

[8]

QUESTION 4

Simplify the following expressions without the use of a calculator:

4.1
$$\frac{\cos(20^\circ - x)\cos x - \sin(20^\circ - x)\sin x}{\sin 215^\circ \cos 325^\circ} \quad (7)$$

4.2
$$2\cos^2 22,5^\circ - \sin^2 48^\circ - \sin^2 42^\circ \quad (4)$$

[11]

QUESTION 5

5.1 Determine the general solution of $2\sin\theta\cos\theta = 2 - 4\sin^2\theta$. (6)

[6]

QUESTION 6

6.1 Prove the identity:
$$\frac{1 - \tan x}{1 + \tan x} = \frac{\cos 2x}{1 + \sin 2x} \quad (7)$$

6.2 Determine the values of x for which the identity above is undefined. (4)

[11]